SOFTWARE TEST PLAN (STP)

Student Attendance Management System (SAMS)

Field	Details
Project	Student Attendance Management System (SAMS)
Version	1.0
Authors	Venkatesh, Vishal, V Sai Sri Saketh, Thejaswini
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Status	Final

Revision History

Version	Date	Author	Change Summary	Approval
1.0	15-09-2026	Project Team Name: Venkatesh SRN:PES2UG24CS580 Name: Vishal SRN:PES2UG24CS591 Name: VUMMADISINGU SAI SRI SAKETH SRN: PES2UG24CS602 Name: Thejaswini SRN: PES2UG24CS563	Initial SAD derived from SAMS SRS	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	Pavithra		

1. Introduction

Purpose: This document defines the software test plan for the Student Attendance Management System (SAMS) v1.0. It outlines the test objectives, scope, strategy, environment, schedule, deliverables, responsibilities, risks, traceability, metrics, and approval activities.

Scope: Testing covers SAMS authentication and role-based access, student/faculty management, courses/sections/timetable, attendance marking through manual entry, QR code and optional RFID/biometric devices, attendance correction and approval, parent/guardian notifications, reports and analytics, leave/on-duty management, dashboards, PDF/Excel export, and audit logging. Optional integrations such as SIS/LMS, SSO/LDAP, SMS and email services are tested at their SAMS integration boundary. Internal implementation of third-party services is outside the SAMS test scope.

References: Student Attendance Management System (SAMS) – Software Requirements Specification (SRS), Version 1.0, dated 07-09-2026; Software Test Plan (STP) – ATM System (Sample) used as the SE test-plan structure/template.

Definitions: SAMS (Student Attendance Management System), SRS (Software Requirements Specification), STP (Software Test Plan), UI (User Interface), API (Application Programming Interface), QR (Quick Response), RFID (Radio Frequency Identification), RBAC (Role-Based Access Control), TLS (Transport Layer Security), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing).

2. Test Items

- Authentication and Role Management module
- Student Management module
- Faculty Management module
- Course, Section and Timetable Management module
- Attendance Management module – manual attendance
- Attendance Management module – QR-code check-in
- Attendance Management module – RFID/biometric capture where institution hardware is installed
- Attendance Correction and Approval workflow
- Parent/Guardian Notification module – SMS/email
- Reports and Analytics module
- Student attendance dashboard
- Leave / On-Duty Management module
- Administration and Configuration module
- Audit Logging module
- Web UI and accessibility functions
- Optional SIS/LMS, SSO/LDAP, SMS and Email integration interfaces

2.1 Role-Based Test Coverage

Role-specific access and workflows shall be tested for all five SAMS roles defined in the SRS.

SAMS Role	Access / Workflow Areas to Test	Representative Test Focus
Administrator	User/account management, configuration, timetable and audit access	RBAC, account lockout/reset, configuration changes, audit logging
Faculty / Teacher	Assigned scheduled attendance, attendance correction request, attendance history	Authorized attendance marking, QR attendance, correction request and role restrictions
Student	Attendance viewing, dashboard, leave/on-	View-only attendance access, dashboard

	duty submission	values, leave/on-duty submission and restrictions
Parent / Guardian	Attendance information and notification receipt	Notification delivery, attendance information access and unauthorized-function restrictions
Academic Coordinator	Attendance correction approval/rejection, academic monitoring	Correction approval workflow, approved correction update and audit preservation

3. Features to be Tested

Features mapped to SAMS SRS requirement IDs:

SRS Requirement ID(s)	Feature to be Tested
SAMS-F-001–F-003	User authentication, account lockout after repeated failures, administrator password reset and forced password change
SAMS-F-004–F-007	Student management and related role-based access
SAMS-F-008–F-010	Faculty management and related role-based access
SAMS-F-011–F-013	Course/section/class and timetable-related management
SAMS-F-014	Manual attendance within the configured post-class window
SAMS-F-015	Attendance authorization and prevention of unauthorized modification
SAMS-F-016–F-017	Attendance percentage calculation, history and status viewing
SAMS-F-020–F-022	Attendance correction request, coordinator approval/rejection, approved correction and audit trail
SAMS-F-030–F-031	Absence and attendance-threshold SMS/email notifications
SAMS-F-040–F-043	Daily/subject-wise/semester reports, defaulter list, PDF/Excel export and student dashboard
SAMS-F-050–F-052	Leave/on-duty submission, approval/rejection and approved-leave treatment
SAMS-F-060–F-062	Administrator management, timetable configuration and audit logging
SAMS-NFR-001	Attendance-screen readiness within 3 seconds for 90% of requests under normal classroom network conditions
SAMS-NFR-002	99.5% availability during institution working hours (8 AM–6 PM), excluding scheduled maintenance
SAMS-NFR-003	Protection of student personal data, encryption at rest, TLS and privacy-policy requirements
SAMS-NFR-004	Attendance and audit-data retention for at least 3 academic years
SAMS-NFR-005	WCAG 2.1 AA accessibility where applicable, including screen-reader support and high-contrast mode
SAMS-SR-001–SR-005	TLS 1.2+, RBAC, password hashing/salting, tamper-evident audit logging and 30-minute inactivity timeout

4. Features Not to be Tested

- Internal implementation of external SIS/LMS systems is outside SAMS scope; only the defined SAMS integration interface is tested where the integration is enabled.
- Internal implementation of SMS/email gateway providers is outside SAMS scope; SAMS notification requests, responses, retry/backoff handling and delivery-status logging are tested at the integration boundary.
- Internal implementation of institution SSO/LDAP is outside SAMS scope; SAMS authentication integration behavior is tested at the defined interface.
- RFID/biometric device firmware and hardware maintenance are outside SAMS scope; SAMS device-event integration is tested where institution-provided hardware is installed.
- Institution core academic/examination systems and payroll/HR systems are outside SAMS scope except for defined integrations.

5. Test Approach / Strategy

Levels

- Unit tests (module-level verification of application components)
- Integration tests (UI/application/database and defined third-party integration boundaries)
- System tests (end-to-end SAMS workflows)
- Acceptance tests (UAT against SRS acceptance criteria)

Types

- Functional testing – verify all SAMS functional requirements and business rules.
- Regression testing – rerun affected tests after defect fixes or changes.
- Performance testing – verify the attendance-screen response target and availability requirement.
- Security testing – verify TLS, RBAC, credential protection, audit logging and session timeout.
- Usability and accessibility testing – verify role-specific UI, screen-reader support and high-contrast mode where applicable.
- Integration testing – verify QR/device, SMS/email and optional SIS/LMS/SSO/LDAP interfaces where enabled.

Entry Criteria

- Stable SAMS build delivered for testing.
- Approved SAMS SRS and corresponding test requirements available.
- Test environment configured and accessible.
- Required test data for students, faculty, courses, sections, timetable and attendance available.
- Required optional hardware or integration test stubs are available when those features are included in the test cycle.

Exit Criteria

- 100% of planned test cases executed.
- All High-priority functional requirements are implemented and verified.
- No critical NFR failures remain open.
- All listed acceptance criteria are satisfied.
- The RTM shows the planned test cases executed with results recorded.
- Defects that prevent acceptance are resolved or formally accepted by the responsible approver.

5.1 Security Validation

- Verify TLS 1.2 or higher is mandatory for client-server and third-party API connections.
- Verify server-side RBAC prevents unauthorized roles from accessing restricted functions and data.
- Verify stored passwords are hashed and salted using an approved algorithm and plaintext passwords are not stored or logged.
- Verify administrative and attendance-modification actions record actor ID, timestamp and before/after values and that audit entries are tamper-evident.
- Verify sessions expire after 30 minutes of inactivity and require re-authentication.
- Verify students cannot modify attendance records.
- Verify sensitive student personal data is protected and attendance data is protected from unauthorized access or modification.

6. Test Environment

Hardware:

- Laptop/desktop/tablet/smartphone using a standard modern web browser.
- Device camera for QR-code check-in testing.
- RFID card reader and/or biometric device where institution hardware is installed and available.

Software:

- SAMS application under test.
- Relational database used by the SAMS implementation.
- Configured SMS/email test gateway or test/stub endpoint where required.
- Optional SIS/LMS and SSO/LDAP test endpoints where those integrations are enabled.

Tools:

- Browser-based UI testing tools as selected by the project team.
- API testing tool for API/integration verification where APIs are exposed.
- Performance/load testing tool for response-time measurement.
- Defect tracking mechanism selected by the project team.

Test Data:

- Dummy administrator, faculty, student, parent/guardian and academic-coordinator accounts.
- Sample courses, sections, faculty assignments and timetables.
- Attendance records covering Present, Absent, Late and approved-leave cases.
- Correction requests with valid/invalid reasons and approval/rejection outcomes.
- Leave/on-duty applications with supporting-document cases where applicable.
- Sample report and notification data.
- Negative/security test data for invalid credentials, unauthorized access and expired sessions.

7. Test Schedule

The exact execution dates are to be finalized by the project team and course coordinator. The sequence follows the SE Test Plan Template.

Milestone / Test Activity	Date / Schedule
Test Planning	TBD
Test Case Preparation	TBD
Test Environment Setup	TBD
Test Execution	TBD
Defect Logging & Tracking	During test execution
Defect Fixing & Retesting	As defects are resolved
Regression Testing	After major fixes / changes
User Acceptance Testing (UAT)	TBD
Final Test Summary Report	After testing and UAT

Note: Exact calendar dates are marked TBD because no project-specific testing dates were provided. The sequence above defines the planned order of testing activities.

8. Test Deliverables

- Software Test Plan (this document)

- Test Cases (manual and/or automated)
- Test Scripts, where automation is used
- Test Data
- Test Execution Logs
- Defect Reports
- Requirements Traceability Matrix (RTM)
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Thejaswini	Prepare plan, coordinate execution
Test Engineer	VUMMADISINGU SAI SRI SAKETH	Design & execute test cases, log defects
Developer	Vishal , Venkatesh	Support defect fixes and triage
Product Owner		Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable SAMS build delivery	Request early test builds and conduct smoke testing before full execution.
Test environment or database unavailable	Maintain a recoverable test environment and backup test data.
Dependency on RFID/biometric hardware	Coordinate hardware availability early; use defined test/stub interfaces where appropriate.
Dependency on SMS/email gateways	Use test gateway/stub endpoints and verify retry/backoff and delivery-status behavior.
Network connectivity problems during classroom testing	Test failure handling and ensure no incomplete attendance transaction is treated as successful.
Incorrect attendance percentage or leave treatment	Use controlled attendance datasets and verify calculations against defined SRS rules.
Unauthorized attendance modification	Execute RBAC and authorization tests at the application boundary, not only through UI controls.

11. Assumptions & Dependencies

- Faculty members have institution-issued credentials for testing.
- Students, faculty, courses, sections and timetable test data are available.
- The relational database required by SAMS is available in the test environment.
- SMS/email gateway dependencies are available or represented by approved test stubs when notification features are tested.

- RFID/biometric features are tested only where institution-provided hardware is installed.
- Optional SIS/LMS and SSO/LDAP integrations are tested only when enabled for the project.

Exact implementation technology, database vendor and hosting platform are implementation details; testing verifies the SAMS requirements at the application and integration boundaries defined by the SRS.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable and blocks meaningful execution.
- The build is too unstable to execute a significant portion of the planned test cases.
- A blocking defect prevents execution of dependent critical workflows.
- Required external dependency or hardware is unavailable for a test scope that cannot be replaced by an approved test stub.

Resume testing if:

- Blocking defects are resolved or an approved workaround is available.
- The test environment is restored and stable.
- Required test data and dependencies are available.
- A new stable build is delivered and smoke testing is completed.

13. Test Case Management & Traceability

The RTM maps SAMS SRS requirements to test cases so that every major requirement has a planned verification path.

Requirement	Test Case Reference
SAMS-F-001	TC-Auth-01
SAMS-F-002	TC-Auth-02
SAMS-F-003	TC-Auth-03
SAMS-F-010	TC-Att-01
SAMS-F-011	TC-Att-02
SAMS-F-012	TC-Att-03
SAMS-F-013	TC-Att-04
SAMS-F-014	TC-Att-05
SAMS-F-020	TC-Corr-01
SAMS-F-021	TC-Corr-02
SAMS-F-022	TC-Corr-03
SAMS-F-030	TC-Notif-01
SAMS-F-031	TC-Notif-02
SAMS-F-040	TC-Rep-01
SAMS-F-041	TC-Rep-02
SAMS-F-042	TC-Rep-03
SAMS-F-043	TC-Rep-04
SAMS-F-050	TC-Leave-01
SAMS-F-051	TC-Leave-02
SAMS-F-052	TC-Leave-03
SAMS-F-060	TC-Admin-01
SAMS-F-061	TC-Admin-02
SAMS-F-062	TC-Admin-03
SAMS-NFR-001	TC-NFR-01
SAMS-NFR-002	TC-NFR-02
SAMS-NFR-003	TC-NFR-03
SAMS-NFR-004	TC-NFR-04
SAMS-NFR-005	TC-NFR-05
SAMS-SR-001	TC-Sec-01

SAMS-SR-002	TC-Sec-02
SAMS-SR-003	TC-Sec-03
SAMS-SR-004	TC-Sec-04
SAMS-SR-005	TC-Sec-05

Additional detailed test cases shall cover the remaining functional requirements in the same requirement groups, including student/faculty management, course/section/timetable management, and all role-specific access paths.

14. Test Metrics & Reporting

Metrics collected:

- % of planned test cases executed
- % of test cases passed, failed and blocked
- Defect count by severity and status
- Defect aging
- Requirement coverage
- Acceptance-criteria coverage
- Performance measurements against the 3-second target
- Availability measurements during the defined 8 AM–6 PM operating period
- Security and accessibility test results

Reports:

- Daily execution status during active testing
- Defect status report
- RTM coverage/status report
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	TBD	
Developer / Dev Lead	TBD	
Course Coordinator / Guide	Pavithra	
Project Team	Venkatesh, Vishal, V Sai Sri Saketh, Thejaswini	

Appendix A – SAMS SRS Coverage Checklist

SRS Area	Covered in STP
Authentication and RBAC	Sections 3, 5.1, 13
Student and Faculty Management	Sections 2, 3, 13
Courses, Sections and Timetable	Sections 2, 3, 6, 13
Manual Attendance	Sections 3, 5, 13
QR Attendance	Sections 3, 5, 6, 13
RFID/Biometric Attendance where hardware exists	Sections 3, 4, 5, 6, 10, 11
Attendance Correction and Coordinator Approval	Sections 3, 5, 13
Parent/Guardian Notifications	Sections 3, 4, 5, 6, 10, 11
Reports, Analytics and Defaulter List	Sections 3, 5, 13
PDF/Excel Export	Sections 3, 5, 13
Student Dashboard	Sections 3, 5, 13
Leave/On-Duty Management	Sections 3, 5, 13
Administration and Audit Logging	Sections 3, 5.1, 13
Performance NFR	Sections 3, 5, 14
Availability NFR	Sections 3, 5, 14
Privacy/Data Protection NFR	Sections 3, 5.1, 14
3 Academic Years Retention NFR	Sections 3, 5, 14
WCAG 2.1 AA NFR	Sections 3, 5, 14
TLS 1.2+, RBAC, password hashing/salting, audit logging, 30-minute timeout	Sections 3, 5.1, 13
Optional SIS/LMS and SSO/LDAP integrations	Sections 3, 4, 6, 10, 11